

SpringDataJpa

Hands-on 4

Difference between JPA, Hibernate and Spring Data JPA

1. Comparison Table

JPA	Hibernate	Spring Data JPA
Specification	ORM Framework	Spring Module
No implementation	Implements JPA	Uses JPA implementation
Defines persistence API	Maps Java objects to DB	Simplifies CRUD operations
Example: @Entity	Example: Session.save()	Example: repository.save()

2. Relationship Diagram

Spring Data JPA

↓

JPA

↓

Hibernate

↓

MySQL

3. Hibernate Code

```
/* Method to CREATE an employee in the database */
public Integer addEmployee(Employee employee){
    Session session = factory.openSession();
    Transaction tx = null;
    Integer employeeID = null;
```

```

try {
    tx = session.beginTransaction();
    employeeID = (Integer) session.save(employee);
    tx.commit();
} catch (HibernateException e) {
    if (tx != null) tx.rollback();
    e.printStackTrace();
} finally {
    session.close();
}
return employeeID;
}

```

4. Spring Data JPA Code

EmployeeRepository.java

```

public interface EmployeeRepository extends JpaRepository<Employee, Integer> {
}

```

EmployeeService.java

```

@Autowired
private EmployeeRepository employeeRepository;

@Transactional
public void addEmployee(Employee employee) {
    employeeRepository.save(employee);
}

```

5. Conclusion

Example:

Spring Data JPA simplifies database operations by reducing boilerplate code. It uses JPA specifications and Hibernate as the default implementation. It provides built-in CRUD operations through the JpaRepository interface and manages transactions automatically.